

- 3 (a) State what is meant by a progressive wave.

.....
 [1]

- (b) A transverse progressive wave travels along a long string PQ from P to Q.

Figure 3.1 shows the string at time $t = 0$.

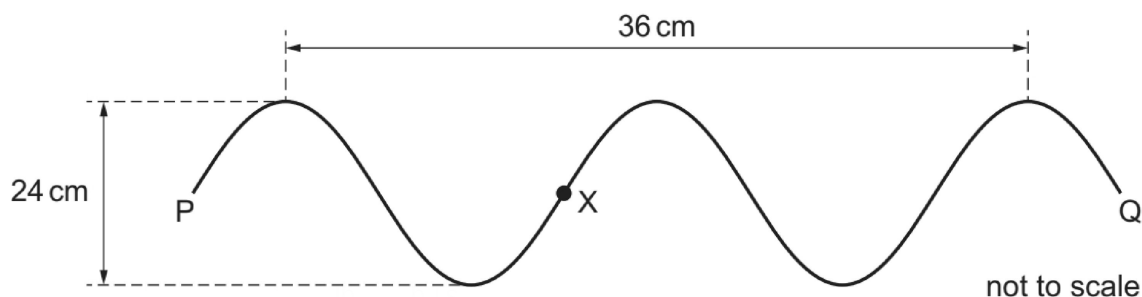


Figure 3.1

A point on the string is labelled X. At $t = 0$, point X is at its equilibrium position. The speed of the wave on the string is 75 cm s^{-1} .

- (i) Show that the period of the wave is 0.24 s.

[2]

- (ii) On Figure 3.2, sketch the variation of the displacement from the equilibrium position of point X with t between $t = 0$ and $t = 0.48 \text{ s}$.

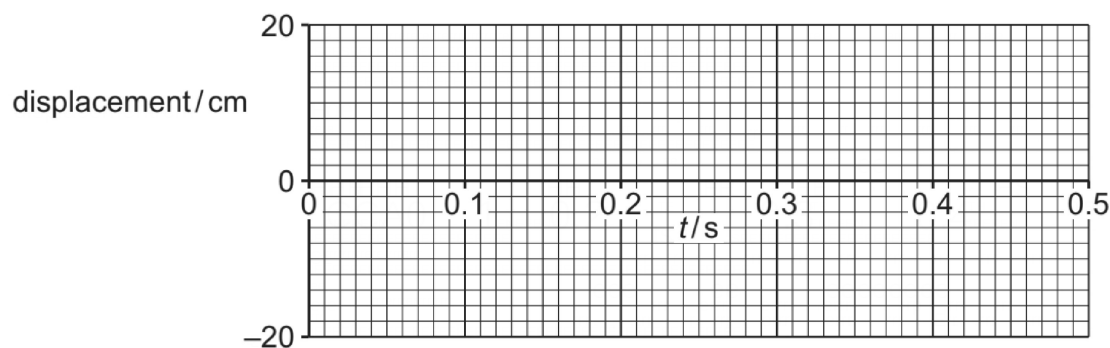


Figure 3.2

[3]

- (c) The string in 3(b) is now fixed to walls at ends P and Q. An oscillator causes transverse oscillations of the string of period 0.24 s. The speed of waves on the string is still 75 cm s^{-1} .

The distance between P and Q is 45 cm.

- (i) Explain how a stationary wave is formed on the string.

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..... [2]

- (ii) Figure 3.3 shows the string at one instant in time.

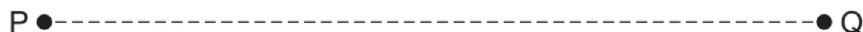


Figure 3.3

On Figure 3.3, draw a solid line to show a possible appearance of the string at a time 0.060 s later. [2]

- (iii) On Figure 3.3, draw:

- a cross (x) labelled with the letter N at **one** point on the string that is at a node
 - a cross (x) labelled with the letter A at **one** point on the string that is at an antinode.
- [2]

[Total: 12]